BASE CHANGE OF DIFFERENTIAL GRADED DIVIDED POWER ALGEBRAS

THE STACKS PROJECT AUTHORS

Editorial note. The closing sentence 'Once we have shown gamma extends it follows easily that this extension has all the desired properties' covers the mechanical transport of the dpdga identities across the tensor product; the load-bearing extension argument is fully displayed.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 1379 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Extension along flat maps. lemma-gamma-extends (same chapter, named). Extends gamma from A_even,+ to A_even (x)_R P along the flat map A_even -> A_even (x)_R P.
- Descent criterion. lemma-kernel criterion (2)(c) (same chapter, named). Certifies descent from A_even (x)_R P to A_even (x)_R R' via generators of the kernel.

Source reference. Original definition 09PJ: sources/stacks-dpa/pinned-original-source.tex, source lines 1300–1300. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 07H0: sources/stacks-dpa/pinned-original-source.tex, source lines 711–711. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07H1: sources/stacks-dpa/pinned-original-source.tex, source lines 720–720. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let (A, d, γ) be as in Definition 09PJ. Let $R \to R'$ be a ring map. Then d and γ induce similar structures on $A' = A \otimes_R R'$ such that (A', d, γ) is as in Definition 09PJ.*

Date: stacks-project:dpa.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Observe that $A'_{even} = A_{even} \otimes_R R'$ and $A'_{even,+} = A_{even,+} \otimes_R R'$. Hence we are trying to show that the divided powers γ extend to A'_{even} (terminology as in Definition 07H0). Once we have shown γ extends it follows easily that this extension has all the desired properties.

Choose a polynomial R-algebra P (on any set of generators) and a surjection of R-algebras $P \to R'$. The ring map $A_{even} \to A_{even} \otimes_R P$ is flat, hence the divided powers γ extend to $A_{even} \otimes_R P$ uniquely by Lemma 07H1. Let $J = \mathrm{Ker}(P \to R')$. To show that γ extends to $A \otimes_R R'$ it suffices to show that $I' = \mathrm{Ker}(A_{even,+} \otimes_R P \to A_{even,+} \otimes_R R')$ is generated by elements z such that $\gamma_n(z) \in I'$ for all $n > 0$. This is clear as I' is generated by elements of the form $x \otimes f$ with $x \in A_{even,+}$ and $f \in \mathrm{Ker}(P \to R')$. $\square$